

Week -6 (June 8 - 11)

Tuesday, 2 June 2026

Fine Tuning -

High level process :

Prepare and upload data



Train and fine tuned model



Evaluate and iterate



Deploy for inference

Tools and Platform for Fine Tuning

Primary platform : OpenAI UI, API & CLI

Data Validation : use OpenAI's data preparation tool to verify JSON format

It always accepts file in JSON format because LLM reads file only this format

Hyper-parameters : (all values are independent)

Batch size -1. Example: book with 100 page, if batch size is 1 , reading 1 page at a time.

Epochs - (4 - 5) standard value. How many times reading the book

Learning rate multiplier - (0.5 - 2) standard value. how fast learning

Data Set - 10

Batch Size - 2 (at once how many data processed) $2 * 5 = 10$

Epochs - 5 [data set- $10 * 5 = 50$] (does not depend on batch size)

Fine Tuning

Fine-tuning means taking a pre-trained AI model and training it further on your own data so it becomes specialized for a particular task, domain, or style.

Real-Time Uses of Fine-Tuning

1. Customer Support Chatbots

A company fine-tunes a model using its past customer support tickets and FAQs.

Example:

User: "How can I return a damaged product?"

Fine-tuned bot: Gives the company's exact return policy and steps.

Benefit: More accurate and company-specific responses.

2. Medical Assistance

Hospitals fine-tune models on medical records, clinical guidelines, and healthcare terminology.

Example:

Doctor enters symptoms.

Model suggests possible diagnoses and relevant medical information.

Benefit: Understands medical language better than a general model.

5. Banking and Finance

Banks fine-tune models on financial terminology and transaction patterns.

Example:

Customer: "Why was my transaction declined?"

Model explains likely reasons and next steps.

Benefit: Faster customer support and financial insights.